Non-Stationarity, Unit Roots and Statistical Inference

Econometrics II

Rasmus Søndergaard Pedersen

Outline

1. Deviations from stationarity:
   - Trends.
   - Level shifts.
   - Variance changes.
   - Unit roots.
2. A general framework for inference in the stationary case.
3. Inference in the non-stationary case.

Stationarity and Weak Dependence

The main assumptions so far has been stationarity and weak dependence.

A time series, y_t, is called *weakly stationary* if

$$\begin{aligned} E(y_t) &= & & \mu \\ V(y_t) &= E((y_t-\mu)^2) & &= \gamma_0 \\ \text{cov}(y_t, y_{t-k}) &= E((y_t-\mu)(y_{t-k}-\mu)) & &= \gamma_k \quad \text{for } k = 1, 2, \ldots \end{aligned}$$

Moreover, y_t is weakly dependent, if y_t and y_{t-k} are approximately independent as $k \longrightarrow \infty$.

Recall that stationarity and weak dependence replace the i.i.d. assumption for the data in cross-sectional regressions and allow applications of the LNN and CLT.

Deviations from stationarity

The stationarity property of a time series can be violated in different ways, e.g.

(A) Deterministic trends (trend stationarity).

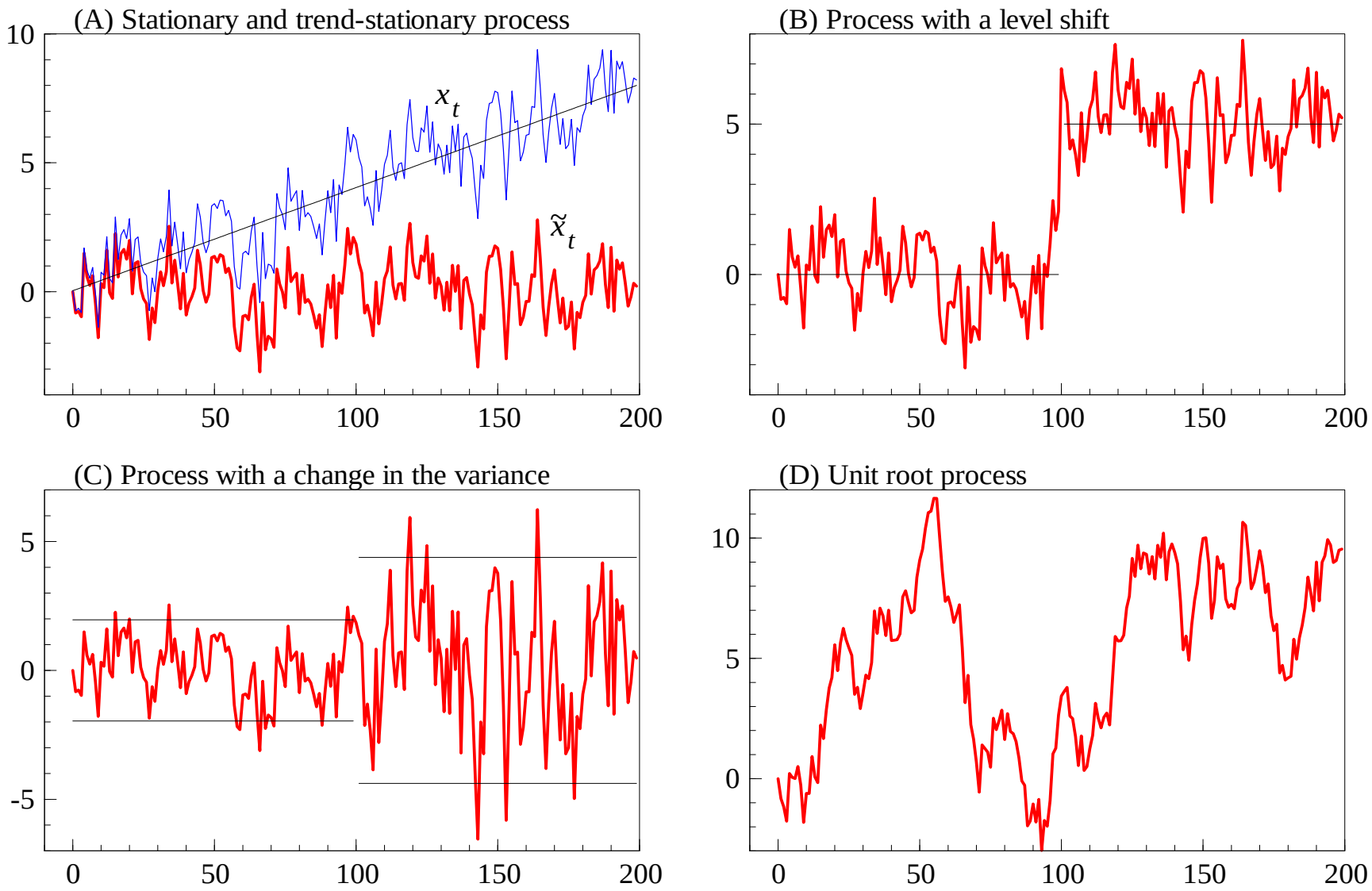
(B) Level shifts.

(C) Variance changes.

(D) Unit roots (stochastic trends).

In terms of statistical inference, each type of deviation must be handled differently.

Four Examples of Non-Stationary Time Series



Case (A): Trend-Stationarity

- Many macro-economic variables are trending.
  How should we model the relationship between trending variables?

- As an example, assume that z_t is stationary

$$z_t = \theta z_{t-1} + \epsilon_t, \quad \text{with} \quad \mid \theta \mid < 1,$$

  and define the process y_t as z_t plus a deterministic linear trend,

$$y_t = z_t + \mu_0 + \mu_1 t.$$

- The process y_t is so-called trend-stationary. Fundamentally, the process is stationary, but its level is trending.

We have that

$$y_t = z_t + \mu_0 + \mu_1 t,$$

with z_t a stationary process with $E[z_t] = 0$.

1. y_t has a trending mean, $E(y_t) = \mu_0 + \mu_1 t$, and is non-stationary.
2. The de-trended variable, $z_t = y_t - \mu_0 - \mu_1 t$, is stationary.
3. We may analyze the OLS-detrended variable, $\hat{z}_t = y_t - \hat{\mu}_0 - \hat{\mu}_1 t$. Standard asymptotics apply to regressions with $\hat{z}_t$.
4. The two-step procedure is equivalent to extending the regression with a deterministic trend, e.g.,

$$y_t = \delta + \theta y_{t-1} + \gamma t + \epsilon_t \quad \text{or} \quad y_t = \beta_0 + \beta_1 x_t + \beta_2 t + \epsilon_t.$$

An example of the Frisch-Waugh-Lovell theorem.

Case (B): Level Shifts and Structural Breaks

- Non-stationarity can result from parameter changes, e.g. a level shift:

$$E(y_t) = \begin{cases} \mu_1 & \text{for} \quad t = 1, 2, ..., T_0 - 1 \\ \mu_2 & \text{for} \quad t = T_0, T_0 + 1, ..., T. \end{cases}$$

If sub-samples are stationary, there are two approaches:

1. Define the relevant shift-dummy

$$D_t = 1(t \geq T_0) = \begin{cases} 0 & \text{for} \quad t = 1, 2, ..., T_0 - 1, \\ 1 & \text{for} \quad t = T_0, T_0 + 1, ..., T, \end{cases}$$

   and include it in the regression model:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 D_t + \epsilon_t.$$

   Again, standard asymptotics apply.
2. Analyze the two sub-samples separately.
   Particularly relevant if we think that more parameters have changed.

Case (C): Changing Variances

- A third type of non-stationary is related to changes in the variance. An example is an AR(1) with

$$y_t = 0.5 \cdot y_{t-1} + \epsilon_t,$$

where

$$\epsilon_t | y_{t-1} \stackrel{d}{=} \begin{cases} N(0,1) & \text{for} \quad t = 1, 2, ..., T_0 - 1 \\ N(0,5) & \text{for} \quad t = T_0, T_0 + 1, ..., T \end{cases}$$

The interpretation is that the time series covers regimes with different volatility.

- A natural solution is to model the regimes separately.

- Alternatively we can try to model the variance.
We return to so-called ARCH models for modelling changing variances later in the course.

Case (D): Unit Roots

- Consider the AR(p) model,

$$y_t = \delta + \theta_1 y_{t-1} + \cdots + \theta_p y_{t-p} + \epsilon_t,$$

with ϵ_t i.i.d.$(0, \sigma^2)$, or,

$$\theta(L) y_t = \delta + \epsilon_t, \quad \text{with} \quad \theta(z) = 1 - \theta_1 z - \cdots - \theta_p z^p.$$

An autoregressive unit root is given by

$$\theta(1) = 1 - \theta_1 - \cdots - \theta_p = 0,$$

or simply $1 = \theta_1 + \cdots + \theta_p$.

- Then the series y_t is non-stationary and behaves as a random walk.
- A simple example is for $p = 1$ and $\theta_1 = 1$, such that

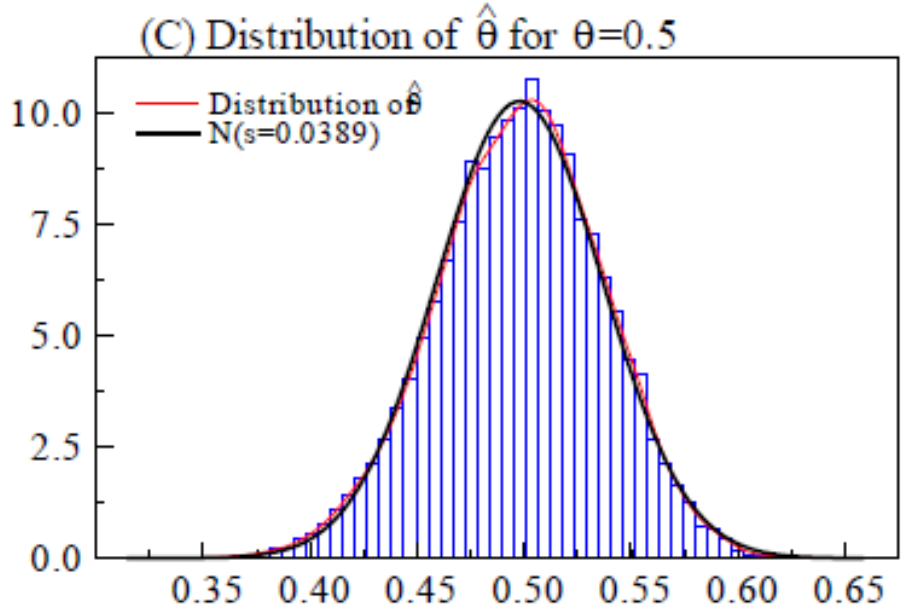
$$y_t = y_{t-1} + \epsilon_t = y_0 + \sum_{i=1}^{t} \epsilon_i.$$

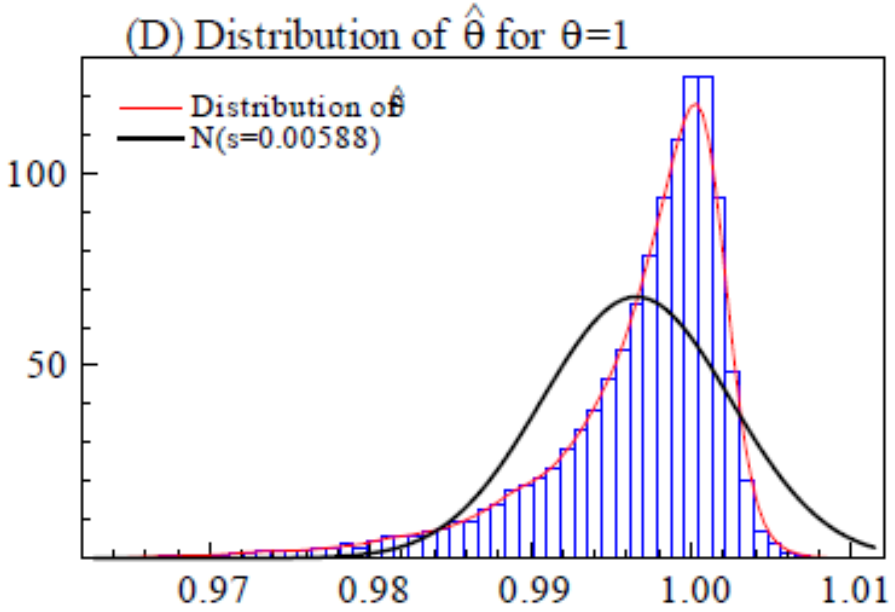
- If the autoregressive polynomial has a unit root, asymptotics change!
  Consider the DGP

$$y_t = \theta y_{t-1} + \epsilon_t, \quad \epsilon_t \mid y_{t-1} \overset{d}{=} N(0,1),$$

  for $t = 1, 2, ..., 500$, and $y_0 = 0$.
  Consider the simulated distribution of $\hat{\theta}$ for $\theta = 0.5$ versus $\theta = 1$.

- Note the shape, location and variance of the distributions.





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The General Dynamic Model

- Let $X_t = (x_{1t}, ..., x_{pt})' \in \mathbb{R}^p$ be variables we want to model.
  Let $W_t = (w_{1t}, ..., w_{mt})' \in \mathbb{R}^m$ be additional explanatory variables.

- Consider a dynamic regression model

$$\begin{aligned} X_t &= \mu + \Pi_1 X_{t-1} + ... + \Pi_k X_{t-k} \\ &\quad + \Psi_0 W_t + \Psi_1 W_{t-1} + ... + \Psi_k W_{t-k} + \epsilon_t, \end{aligned} \qquad (\bigstar)$$

  where ϵ_t is i.i.d.$(0, \Omega)$.

- The model has $n = p + k \cdot p^2 + k \cdot p \cdot m$ parameters

$$\theta = \{\mu, \Pi_1, ..., \Pi_k, \Psi_0, \Psi_1, ..., \Psi_k\}.$$

  Special cases include AR, ADL, VAR, and VAR-X.

Inference in the Stationary Case

- So far we have used standard LLN and CLT.

- If (X_t, W_t) is stationary and weakly dependent, and the model is correct, then for any $j = 1, 2, ..., n$:

$$\sqrt{T}(\hat{\theta}_j - \theta_j) \xrightarrow{d} N(0, V_j),$$

  where V_j is the asymptotic variance. We use the asymptotic approximation

$$\hat{\theta}_j \overset{a}{\sim} N\left(\theta_j, \frac{1}{T} V_j\right)$$

- Convergence holds jointly for all parameters, and joint tests are χ^2.
  1. Lag-length determination in a univariate autoregression.
  2. Lag-length determination in a vector autoregression.
  3. Test for Granger non-causality in a vector autoregression.
- **Note:** Assumption that ϵ_t is i.i.d. can be relaxed. E.g., ϵ_t may be uncorrelated with the regressors but heteroskedastic.

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Non-Stationary Case

Stationarity and weak dependence are sufficient but not necessary conditions. If some variables in X_t or W_t are unit root non-stationary, $I(1)$, result by Sims, Stock and Watson (1990):

If the model ($\bigstar$) can be reformulated in such a way that θ_j, for some $j = 1, 2, ..., n$, is a parameter to a stationary mean zero regressor, then it holds that

$$\sqrt{T}(\hat{\theta}_j - \theta_j) \xrightarrow{d} N(0, V_j),$$

and tests on θ_j will have standard $N(0, 1)$ and $\chi^2(1)$ distributions.

If two or more parameters, θ_j and θ_h, are parameters to stationary mean zero regressors in the same reformulation of the model, then they are jointly Gaussian and a joint test on θ_j and θ_h will have a standard χ^2-distribution.

Example I: Individual Test

- Let $y_t \in \mathbb{R}$ be unit-root non-stationary and consider

$$y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \theta_3 y_{t-3} + \epsilon_t, \qquad (*)$$

  with $\theta(1) = 1 - \theta_1 - \theta_2 - \theta_3 = 0$.
  Not clear how to perform tests on $(\theta_1, \theta_2, \theta_3)$.

- We can reparametrize the model as

$$y_t = \theta_1 y_{t-1} + (\theta_2 + \theta_3) y_{t-2} + \theta_3 \left(y_{t-3} - y_{t-2}\right) + \epsilon_t. \qquad (**)$$

  Now θ_3 is coefficient to $y_{t-3} - y_{t-2} = -\Delta y_{t-2}$ and $\hat{\theta}_3$ is asymptotically Gaussian.

- We do not have to estimate the reformulated version $(**)$.
  If it exists, we can make the test in $(*)$.

Example II: Joint Test

- We can also write the model as e.g.

$$y_t = \theta_1(y_{t-1} - y_{t-2}) + (\theta_1 + \theta_2 + \theta_3)y_{t-2} + \theta_3\,(y_{t-3} - y_{t-2}) + \epsilon_t.$$

- Here (θ_1, θ_3) are regressors to stationary terms in the same reformulation.

- A joint test on (θ_1, θ_3) is $\chi^2(2)$.
- All of this also works in VAR models.

Conclusion:

- The implication of the above results is that we can do, e.g., lag-length determination in a VAR or AR even in the presence of unit roots.

Example III: Some Non-Standard Situations

Not all situations can be dealt with in this way.
Some important cases:

1. We cannot rewrite the model such that all θ_1, θ_2, and θ_3 are coefficients to stationary regressors at the same time.
   We need at least one term in the model.
2. In a non-stationary VAR models, tests for Granger non-causality are non-standard (they involve all coefficient matrices).
3. We cannot write $\theta_1 + \theta_2 + \theta_3$ as coefficient to a stationary regressor.
   Unit root tests have non-standard distributions.